

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Specification for Pipes and Fittings

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Revision History

Rev No.	Date	Description of Revision
R01	24-05-22	First Issue for Client Review
A01	20.12.22	Issued for Approval.

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- MPP-SPC-010 MPP-SPC-010

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Classes
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

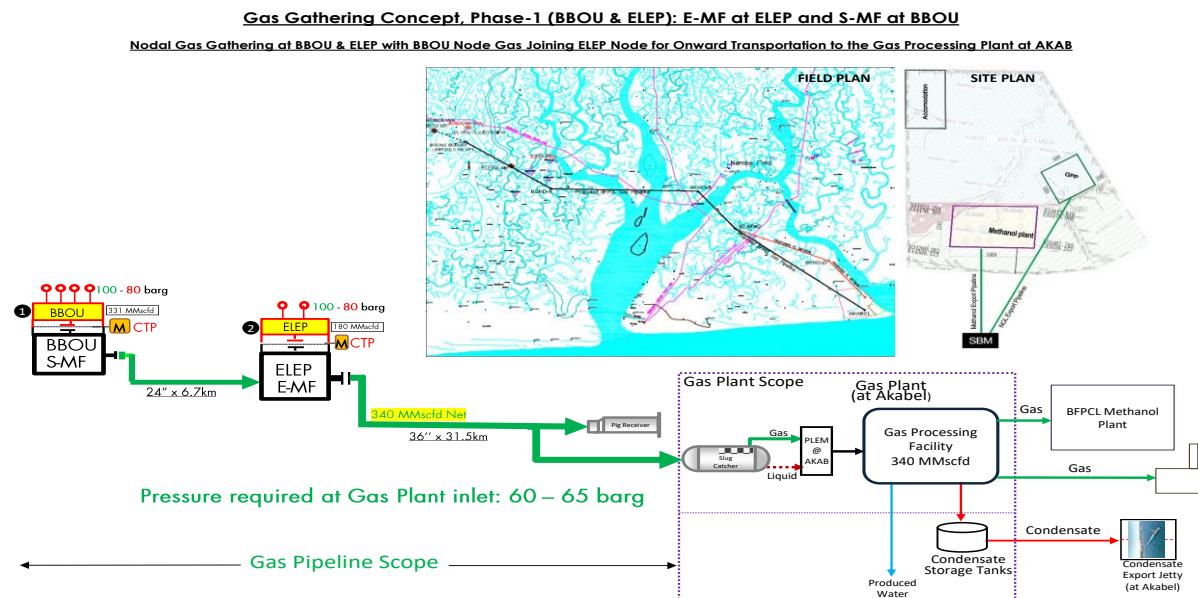


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification applies to metallic pipes and fittings made of carbon steel and Stainless Steel used for in plot piping (as per ASME B31.3) and details key requirements for their design, fabrication, inspection, and supply for BFPCL Phase 1 Project.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.38.01.25-Gen.	Piping – Process Design Requirements
Dep 31.38.01.10-Gen	Piping Classes – Basis of Design
DEP 31.38.01.24-Gen.	Piping – Engineering and Layout Requirements
DEP 31.38.01.31-Gen	Shop And Field Fabrication of Piping
DEP 31.38.01.84-Gen	Piping Classes
DEP 31.38.01.11-Gen	Piping Classes – Refining and Chemicals
MESC SPE 76/210	Carbon Steel Forgings (Amendments/Supplements To ASTM A 105)
MESC SPE 76/101	Flanges (Amendments/Supplements to ASME B16.47)
MESC SPE 76/212	Alloy And Stainless Steel Forgings (Amendments/Supplements To ASTM A 182)
MESC SPE 76/100	Flanges (Amendments/Supplements to ASME B16.5)
ASME B1.1	Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B1.20.1	Pipe Threads, General Purpose
ASME B16.5	Pipe Flanges and Flanged Fittings
ASME B16.9	Factory Made Wrought Steel Butt Welding Fittings
ASME B16.11	Forged Steel Fittings, Socket Welding and Threaded
ASME B16.25	Butt Welding Ends
ASME B31.3	Process Piping
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
ASTM A106	Seamless Carbon Steel Pipe for High Temperature Service
ASTM A234	Pipe Fittings for Wrought Carbon Steel and Alloy Steel for Moderate and Elevated Temperatures
ASTM A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
MSS SP-83	Class 3000# Steel Pipe Unions Socket Welding and Threaded
MSS SP-75	Specification for High-Test, Wrought, Butt-Welding Fittings
MSS SP-97	Integrally Reinforced Forged Branch Outlet Fittings-Socket Welding, Threaded, and Butt Welding Ends

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to.

The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. General Requirements

- Unless otherwise stated in the data sheets, all Pipes and Fittings shall be supplied in the Seamless Condition.
- Welding end preparation shall be in accordance with ASME B16.25
- The corrosion allowance for carbon steel pipes shall be 3 and 6 mm as shown on the data sheets.

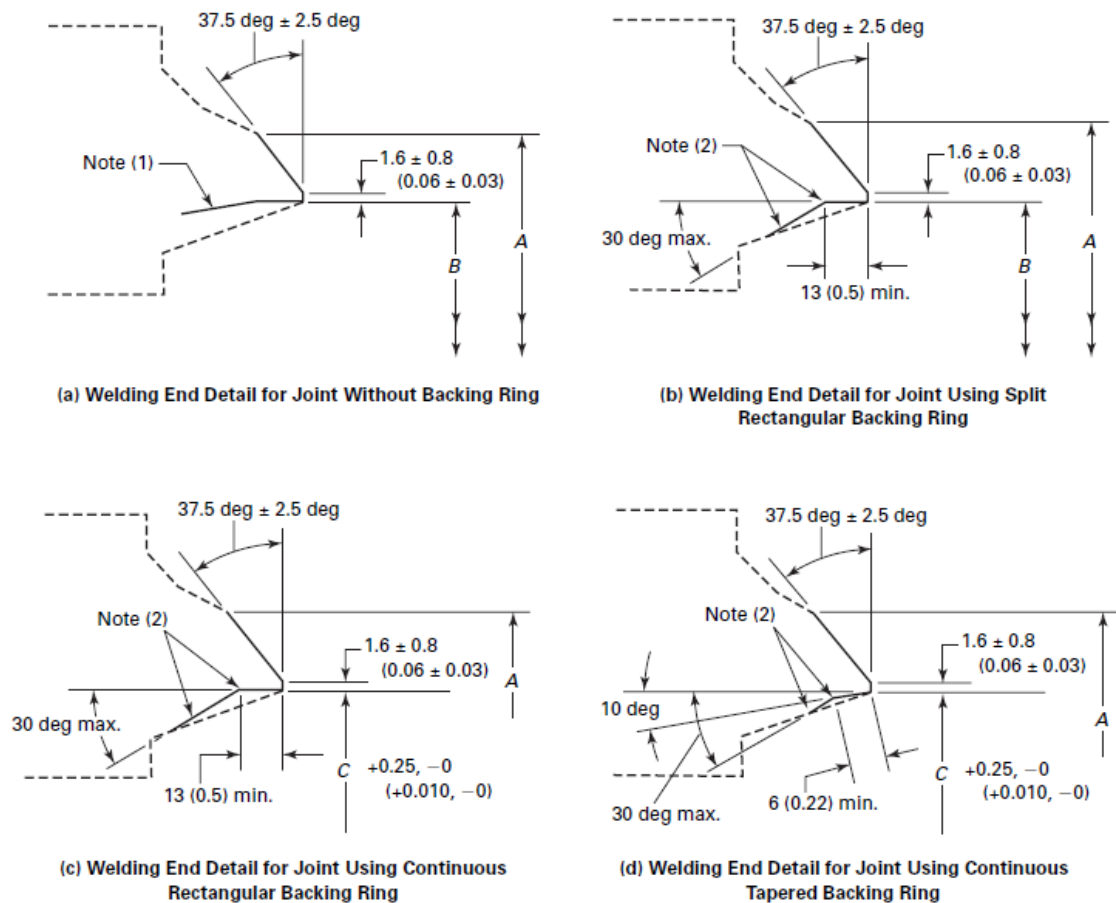
3.1. Pipes

- Metallic pipe shall be in accordance with ASME B36.10M
- Material specification for pipes shall be as per ASTM A106 and /or API 5L
- Stainless Steel Pipes shall be Seamless as per, ASTM A312 TP 316L/ASTM A358 Grade 316L CL1
- Pipes less than 2" in diameter shall have plain ends.
- Pipes 2" and above in diameter shall have bevelled ends for butt-welding connection.
- The minimum wall thickness for pipes 2" and below shall be schedule 80 as per ASME B36.10
- The standard pipe lengths shall not be less than:
 - ✓ 5.33m for diameter up to 4"
 - ✓ 10.51m for diameter 6" and above.

3.2. End Preparation for Pipes

Pipes 2" and above having a wall thickness not more than 22mm shall have single bevel for but welding. The internal and external surface preparation shall follow the details of ASME B16.25. Fig 2. The bevel angle shall be $37.5^{\circ} \pm 2.5^{\circ}$ with a root face dimension of $1.6\text{mm} \pm 0.8\text{mm}$.

Pipes less than 2" shall have plain ends to match socket welding connections in accordance with ASME B16.11

Fig. 2 Bevels for Wall Thickness Over 3 mm (0.12 in.) to 22 mm (0.88 in.), Inclusive**GENERAL NOTES:**

- (a) Broken lines denote maximum envelope for transitions from welding bevel and root face into body of component. See Fig. 1 for details.
- (b) See section 5 for tolerances other than those given in these illustrations.
- (c) Purchase order must specify contour of any backing ring to be used.
- (d) Linear dimensions are in millimeters with inch values in parentheses.

NOTES:

- (1) Internal surface may be as-formed or machined for dimension *B* at root face. Contour within the envelope shall be in accordance with section 2.
- (2) Intersections should be slightly rounded.

3.3. Wall Thicknesses

Thicknesses of pipes and fittings shall be in accordance with API 5L or ASME B36.10M. A negative mill tolerance of 12.5% shall be applied for all seamless pipes as required in ASME B31.3.

When wall thicknesses calculated is not listed within ASME B36.10M, in such cases the wall thickness calculated shall be the minimum thickness (mm) required. The next available standard wall thickness listed in the above-mentioned standards shall be selected for procurement.

If it is imperative that the Bidder proposes an alternative wall thickness perhaps due to manufacturing constrains, it shall never be less than the calculated wall thickness, then all piping, fittings and flanges of the same diameter and nominated thickness shall be increased to the alternative

thickness proposed by the Bidder.

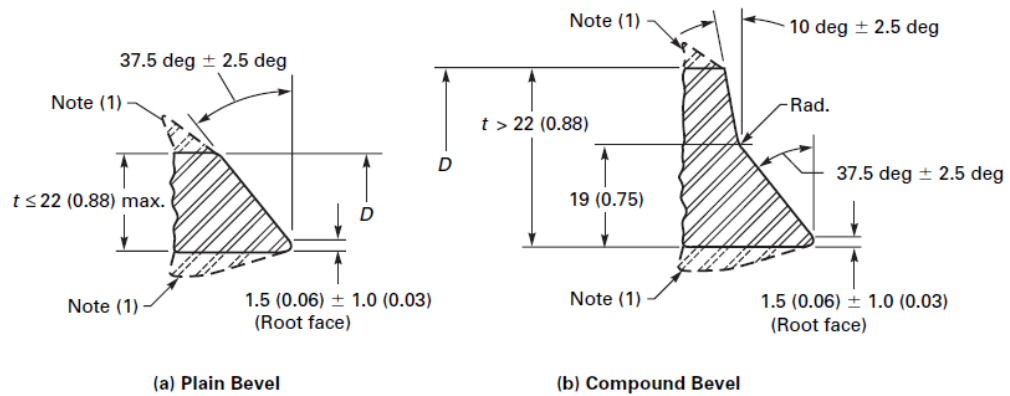
3.4. Fittings

- Fittings 2" and above shall be made of the butt-welding type. Butt-welding fittings shall be in accordance with ASME B16.9.
- Dimensions of flanged fittings shall generally be in accordance with ASME B16.5.
- As a general rule, the fittings used for piping up to 1 ½" in diameter shall be made of forged steel, of the socket-welding type. Forged fittings shall be forged close to final shape before machining to ensure adequate grain flow at changes of section.
- Supply of heavier wall thickness fittings taper bored down to the required thickness is not permitted, without specific written approval from the MANUFACTURER and Purchaser in each case.
- Sockolet, Weldolet and Thredolet and other similar fittings shall be design and manufactured in accordance ASME B16.11 and with MSS SP-97. Dimensions shall comply with ASME B16.11 and with MSS SP-97.
- Flangeolets if applicable shall have an overall length of 150mm, unless otherwise agreed
- For fittings with dimension different from those specified in ASME B16.9 and MSS SP 43, vendor must submit detailed document showing all dimensions, calculations and tolerances of fittings in accordance with ASME B31.3 for EMPLOYER and MANUFACTURER for approval.
- Elbows 2" and above shall have a large radius ($R=1.5 D$).
- Reducing elbows, straight crosses and reducing outlet crosses are regarded as special fittings and should be avoided

3.5. End Preparation for Fittings

Fittings 2" and above having wall thickness of not more than 22mm shall have plain bevel as illustrated in table 12 of ASME B16.9 (See Sketch below). Bevelling may be performed by machining or by flame cutting. The process used shall produce clean, square, burr-free cuts. To achieve a 75° angles V grooves for weld deposit, fitting shall have a standard bevel angle of $37.5^\circ \pm 2.5^\circ$ with a root face dimension of $1.5\text{mm} \pm 1.0$.

Fittings less than 2" shall be furnished with a bell end in accordance with ASME B16.11 standards prepared for socket welding connection

Table 12 Welding Bevels and Root Face

Nominal Wall Thickness, t	End Preparation
Less than x [Note (2)]	Cut square or slightly chamfer, at manufacturer's option (not illustrated)
x to 22 (0.88), inclusive	Plain bevel as in illustration (a) above
More than 22 (0.88)	Compound bevel as in illustration (b) above

GENERAL NOTES:

- (a) Dimensions in parentheses are in inches.
 (b) Other dimensions are in millimeters.

NOTES:

- (1) See section 8 and Fig. 1 for transition contours.
 (2) $x = 5 (0.19)$ for carbon steel or ferritic alloy steel and 3 (0.12) for austenitic alloy steel.

Specific Material Requirements

3.6. Carbon Steel and LTCS Piping, Fittings

The carbon steel for the fabrication of seamless pipes and fittings shall be produced in an electric furnace or by the basic oxygen process.

By product analysis (ladle Analysis), carbon equivalent shall not be more than 0.43 calculated using the formula:

$$\text{C.E.} = \text{C} + \frac{\text{Mn}}{6} + \frac{\text{Cr} + \text{Mo} + \text{V}}{5} + \frac{\text{Ni} + \text{Cu}}{15}$$

Carbon steels, including normalized steels shall have a maximum carbon content of 0.23%.

If the content of alloying elements other than C or Mn is unknown, the formula to be used shall be:

$$\text{C.E.} = \text{C} + \frac{\text{Mn}}{6}$$

In this case, the carbon equivalent shall not be more than 0.40.

All carbon steel materials must be impact tested at minus 29°C (-20°F) in accordance with ASME B31.3 (paragraph 323.3), otherwise Impact-tested steel SHALL be used for carbon steel piping which lower design temperature is below 0 °C (32 °F).

However, carbon steels to API 5L Gr.B or to ASTM A106 Gr.B shall not be tested if their nominal wall thickness is not above 12.7 mm (½ inch).

All carbon steel materials for services at lower temperatures (i.e., below minus 29°C) shall undergo toughness testing in accordance with the requirements of their reference codes and standards.

The material composition of the stainless shall conform to the requirements of A182 F316L and ASTM A403 or WP316LS.

3.7. Quality Assurance and Inspection

Vendor is responsible for all QA/QC services required to ensure strict compliance with technical requirements

Vendor shall prepare and submit Vendor's ITP and Quality plan in tabular form for MANUFACTURER/EMPLOYER approval.

3.8. Certificate of Conformity

Vendor shall submit certificates of conformity with ASTM standards specified in the requisition for all pipes, fittings and flanges.

3.9. Material Certificates

The Vendor shall supply the following material certificates giving the results of the chemical analysis and of the mechanical tests carried out in accordance with the requirements of the referenced codes and standards. Vendor shall submit the following;

- EN 10204 3.1 Carbon Steels, Low Temp Carbon Steel and Stainless Steel
- EN 10204 3.1 Fittings – General

3.10. Preservation and Packing

Pipes and fittings shall be marked by cold stamping with the following information

- End connection nominal size (NPS)
- The specification and grade of material, schedule number or wall thickness
- The heat number
- The batch number
- Diameter

In Addition fittings the marking requirements of MSS SP 25 are also applicable.

3.11. Packing

Pipes, fittings and flanges must be packed in sea-packing; they must meet the conditions for all stages of transport as applicable in this project (road, air and sea). Each unit shall be completely dried prior to shipment Vendor shall be responsible for all materials damage that occurs as a result of improper drainage or packing.

If machined exterior surfaces shall be protected from corrosion during shipment and subsequent storage. Bevelled fitting ends shall be protected by a peel-off varnish

The packing list shall clearly show as a minimum;

- Purchase order number
- Nominal Size/Thickness
- Material description
- Heat number
- Batch number
- Colour code
- Package weight

3.12. Documentation

The Vendor's documentation and all certificates shall comply with the following:

- It shall be in English language only.
- It shall be in SI units, although flange diameters shall be in nominal inch size.
- Drawings and other documentation shall be in A3 and/or A4 size as appropriate.